TEXORA · LECTURE SHEET

Sub-station: Purpose, Types and Equipment

Why substations exist in the power system; how they are classified; and the equipment found in one — HT and LT switchgear, current and potential transformers, the step-down transformer, distribution boards and sub-distribution boards — read through a single-line diagram.

EEE-2203 · BASICS OF ELECTRICAL & ELECTRONICS ENGINEERING · TOPIC 3 · LEVEL 2 TERM 2

1. Purpose of a sub-station

A **sub-station** is the part of a power system where voltage is transformed from one level to another and where circuits are switched, protected, measured and controlled. Power is generated and transmitted at high voltage (for example 132 kV or 230 kV) to keep current — and therefore line losses — low, but factories and homes need a few hundred volts. The substation bridges the two. Its main purposes are:

- **Voltage transformation:** stepping voltage up for transmission or down for distribution and use (for a textile mill, typically from an 11 kV utility supply to 415 V three-phase for machinery).
- **Switching:** connecting or disconnecting feeders, transformers and busbar sections for operation, maintenance and fault isolation.
- **Protection:** housing the relays and circuit breakers that isolate a faulty section automatically before damage spreads.
- **Measurement and control:** metering energy, current and voltage, and providing for power-factor correction and voltage regulation.
- **Safety:** earthing, clearances, interlocks and fencing that keep people away from live high-voltage parts.

2. Types of sub-station

CLASSIFICATION	TYPES	NOTES
By function	Step-up (transmission) sub-station	At the power station; raises generator voltage (e.g. 11 kV to 132 kV) for bulk transmission.

	Step-down (distribution) sub-station	Lowers transmission or sub-transmission voltage (132/33 kV, 33/11 kV) towards the consumer.
	Consumer / industrial sub-station	Owned by a large consumer such as a factory; takes an 11 kV supply and distributes 415 V inside the plant.
By construction	Outdoor sub-station	Equipment in the open air on steel structures; used for higher voltages, cheaper, needs land and fencing.
	Indoor sub-station	Equipment inside a building; used up to about 33 kV where land is costly or pollution is severe.
	Pole-mounted sub-station	A small transformer (up to about 250 kVA) on poles; the common rural and small-town distribution unit.
	Foundation / plinth-mounted and compact (package) sub-stations	Ground-mounted units; the package type arrives factory-assembled with transformer and switchgear in one enclosure.
By service	Converter, traction, mobile sub-stations	Special-purpose: A.C.–D.C. conversion, railway supply, or a trailer-mounted emergency unit.

3. The single-line diagram

Substations are drawn as **single-line diagrams**: one line represents all three phases, and each item of equipment has a standard symbol. The diagram below is the skeleton of a small industrial substation. Follow the energy: it arrives on the 11 kV feeder (top), passes protection and switching equipment to the busbar, is stepped down by the transformer, and leaves through the LT board to the factory’s feeders.

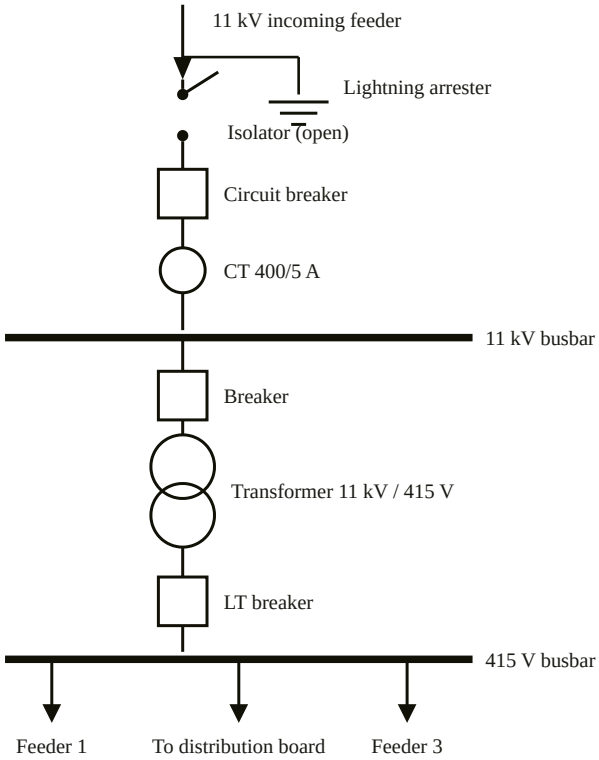


Fig. 1 — Single-line diagram of a small industrial sub-station: 11 kV in, stepped down to 415 V, out to the works feeders.

4. Sub-station equipment and what each item does

EQUIPMENT	FUNCTION
Power transformer	The heart of the substation: steps the voltage down (or up) by electromagnetic induction (Topic 2). Fitted with radiators or fans for cooling, a conservator for the oil, and taps to adjust the output voltage.
Busbar	A heavy copper or aluminium bar (or tube) to which incoming and outgoing circuits connect; the substation’s common node. Sections can be coupled or split by bus couplers.
HT switchgear	The high-tension (e.g. 11 kV) switching equipment: circuit breakers, isolators and fuses in metal-clad or open form, rated to make and break load and fault currents at HT.
LT switchgear	The low-tension (415 V) equivalent: air circuit breakers, MCCBs and switch-fuse units that feed and protect the outgoing LT circuits.
Circuit breaker	Switches a circuit under load and, critically, breaks fault current automatically on a signal from its relay (details in Topic 4).
Isolator (disconnect switch)	An off-load switch giving a visible open gap for safe maintenance; never used to break load current.

Current transformer (CT)	Steps a large primary current down to a standard 5 A (or 1 A) secondary for meters and protective relays, and insulates them from the HT line. A 400/5 CT has a ratio of 80 : 1. The secondary must never be left open-circuited while the primary carries current — a dangerously high voltage would be induced.
Potential transformer (PT)	Steps a high voltage down to a standard 110 V secondary for meters and relays (e.g. 11 000/110 V, ratio 100 : 1), again providing insulation from the HT system.
Lightning arrester	Connected between line and earth; it offers a low-resistance path for lightning and switching surges, discharging them to earth while blocking normal power-frequency current.
Distribution board (DB)	The main panel at the end of a feeder: it splits the supply into sub-circuits, each with its own protective device (MCB, MCCB or fuse), and feeds the sub-distribution boards.
Sub-distribution board (SDB)	A smaller board on each floor or section, fed from a distribution board, which in turn feeds the individual machines, lighting and socket circuits. The DB → SDB → final circuit chain keeps faults local and maintenance safe.
Earthing system	Earth electrodes, earth bus and connections that hold exposed metalwork at earth potential and carry fault current (Topic 4).

5. Metering through instrument transformers

Meters and relays are never connected directly to an 11 kV line. The CT and PT scale the quantities down, and readings are multiplied back by the instrument-transformer ratio:

$$\text{Primary current} = \text{CT ratio} \times \text{ammeter reading} \quad \cdot \quad \text{Primary voltage} = \text{PT ratio} \times \text{voltmeter reading}$$

WORKED EXAMPLE — READING A CT AND A PT

A CT is rated 200/5 A, so its ratio is 200/5 = 40. If the ammeter on its secondary reads 3.2 A, the primary current is 3.2 × 40 = **128 A**.

A PT is rated 11 000/110 V, ratio 100. If its voltmeter reads 85 V, the primary voltage is 85 × 100 = 8500 V = **8.5 kV** — the line is running below its 11 kV rating.

6. Key terms

- **Sub-station:** an installation where voltage is transformed and circuits are switched, protected and metered.
- **Busbar:** the common heavy conductor to which the circuits of a substation connect.
- **Switchgear:** the combination of switches, circuit breakers, fuses and their controls used to switch and protect circuits; HT for the high-voltage side, LT for the low-voltage side.
- **Feeder:** a circuit that carries power from a busbar to a load or a distribution point, without intermediate taps.
- **Instrument transformer:** a CT or PT that scales current or voltage down to standard meter values and insulates the instruments from the power circuit.
- **Single-line diagram:** a diagram in which one line represents the three phases of an A.C. system.
- **Distribution board / sub-distribution board:** panels that split a supply into progressively smaller protected circuits.

7. Key points

- A substation exists to transform voltage and to switch, protect, measure and control power at a node of the network.
- Classification is by function (step-up, step-down, consumer), by construction (outdoor, indoor, pole-mounted, package) and by special service.
- Energy flows: feeder → lightning arrester, isolator, breaker → HT busbar → transformer → LT breaker → LT busbar → feeders → distribution board → sub-distribution boards → final circuits.
- CTs and PTs make HT metering safe and standard: multiply the meter reading by the ratio to get the primary value.
- Isolators give a visible safety gap but must never break load current; only breakers are rated to interrupt current.

8. Summary

The sub-station is the controlled junction between the high-voltage supply network and the low-voltage world of the factory floor. Its transformer changes the voltage level; its busbars organise the connections; its HT and LT switchgear switch and protect; its CTs and PTs let ordinary instruments supervise a high-voltage system; and its chain of distribution and sub-distribution boards delivers power to the final circuits in protected, maintainable sections. Read any substation as a single-line diagram and the whole installation becomes a story of energy flowing downhill in voltage, under protection at every step.

9. Practice

Answer in your own words or by calculation; brief answers follow. No multiple-choice questions.

1. An ammeter on the secondary of a 400/5 A CT reads 4.1 A. What is the primary current?
2. A factory receives power at 11 kV but its machines run at 415 V. Sketch the sequence of equipment the energy passes through, naming each item.
3. Why is a CT secondary never left open-circuited while the primary is energised?
4. Distinguish between a circuit breaker and an isolator in two sentences.
5. Give two reasons why a large factory prefers its own indoor consumer substation to a pole-mounted one.

Answer key

1. Ratio = 400/5 = 80. Primary current = 4.1 × 80 = 328 A.
2. 11 kV feeder → lightning arrester (to earth) → isolator → HT circuit breaker → CT (for metering/protection) → 11 kV busbar → breaker → step-down transformer (11 kV/415 V) → LT circuit breaker → 415 V busbar → outgoing feeders → distribution board → sub-distribution boards → machines.
3. The CT primary current is fixed by the load, not by the secondary. With the secondary open, no opposing secondary ampere-turns exist, so the whole primary current magnetises the core; the core saturates and induces a very high, dangerous voltage across the open secondary terminals, endangering staff and insulation.
4. A circuit breaker can make and break load and fault currents safely and operates automatically under relay control. An isolator is an off-load device that only provides a visible isolating gap after the breaker has opened the circuit.
5. Capacity: a pole-mounted transformer is limited to a few hundred kVA, below a large factory’s load. Reliability and safety: indoor switchgear is protected from weather and pollution, can include full relay protection and metering, and can be maintained under cover without outage risks to the public supply.